GEOGRAPHY AND REPRESENTATION

The splendid sunsets of today's massively multiplayer online role-playing games seem only distantly related to the primitive (if groundbreaking) two-color visuals of *Spacewar!* This section provides an account of the various strategies of geography and representation employed by game designers.

Obviously, the way the game world is laid out is related to the way it looks. If you're roaming a three-dimensional city and the way ahead is blocked, you will typically notice something large in front of you (as opposed to just suddenly being stopped by an invisible barrier). But there is also a difference between geography and representation. In the 1982 arcade game *Moon Patrol*, the player patrols a lunar surface for aliens, while futuristic buildings scroll by in the background. These buildings have no direct in-game effect; for instance, they don't block the player's movement or provide cover. They are simply there for the visual effect. Similarly, the "splendid sunsets" referred to above have no direct bearing on the player's possibilities. They mostly add to the atmosphere, provide a sense of realism, and generally make the game world seem alive.

Game students and scholars who focus closely on rules run the risk of reducing audiovisuals to the category of surface phenomena. Graphics, in particular, are sometimes treated as mere window dressing, eye candy providing an enticing way of interacting with the actual game beneath. As we shall see, that view is much too simple, as graphic types have different properties and afford different gameplay styles. Also, a graphical style may ideally be chosen for its ability to support the game mechanics, but in the real world the causal arrow is sometimes reversed, as graphics determine the mechanics. For instance, if a game designer starts out with a preference for a certain graphical style, this preference is likely to influence the kind of game she will make.

Imagine, for instance, a game designer who has access to graphical tools developed for a first-person shooter. All other things being equal, she is unlikely to produce a real-time strategy game. Similarly, if she sets out to make a real-time strategy game, she may debate whether to incorporate 2D or 3D graphics, but she is likely to choose a third-person perspective.

Geography, representation, and gameplay are interrelated. To see this more clearly, let us look at the way in which games have developed over the years. Concerns over the relative merits of 2D and 3D game engines were not pressing on the minds of the *Spacewar!* developers in the early 1960s. Working under severe hardware constraints, they were forced to limit the number of on-screen objects and to limit the details of these objects. Nothing, however, forced them to create the game's puzzling wraparound space, which ensured that a spaceship leaving the right edge of the screen would materialize on the left. It would be odd, yes, to have the ships crash against the screen edges (space is supposed to be infinite, after all), but there is

certainly no logic demanding that the ships make this odd and unexplained transition to another position in space. Although hardware was important, this was one of many design choices, with profound implications for the gameplay of this and many other games to follow.

We see the action of *Spacewar!* from an abstract position—rather than from the perspective of one of the game's protagonists—so we can say that the game employs a third-person perspective. The alternative to this would be a first-person perspective, in which the player sees the game from her character's perspective, as is the case in first-person shooters.

The galactic shoot-out takes place on a single plane, in which the spaceships can move in any of the cardinal directions but not down or up away from the plane. This is equivalent to a space battle carried out on a game board—lacking any depth—and is of course not realistic in any way. Thus, we say that the graphics are two-dimensional. The action takes place on a simple coordinate plane and the position of an object can be described using only two coordinates (see Figure 5.3). This is as opposed to three-dimensional graphics, which also have depth, much like the physical world.

Spacewar! was a competitive two-player game where the players (typically) were watching the same screen; thus it would be difficult to have the perspective move to follow one of the players. The *Spacewar!* world therefore did not scroll. "Scroll" refers to the movement (usually horizontal) of the perspective to follow the action, somewhat like the gradual unrolling of the contents of a paper scroll, or a camera on a track that rolls along beside a scene.

Multiplayer one-display games that scroll have typically been cooperative action games like *Golden Axe, Gauntlet*, or *Double Dragon*, in which it is to players' advantage to move in synchrony. Thus, *Spacewar!*'s nonscrolling perspective may have been a function of gameplay details but another choice made by the designers, one not required by the logic of the game or its technological constraints, is the game's off-screen space. Because the playing field wraps around to meet itself, *Spacewar!* technically has no off-screen space. Thus, the players can see all the objects in the game at all times, and do not have to worry about anything lying beyond the screen. Again, hardware limitations might have played a role, but this design choice also suggests the pervasive influence of classical board games. In chess, *Stratego* and *Risk*, for example, nothing exists beyond the board.

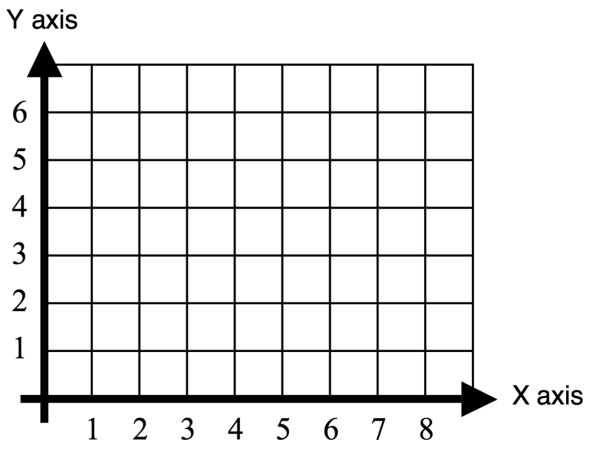


Figure 5.3 Two-dimensional graphics mean that the game action is limited to a simple coordinate system

Though *Spacewar!* is simple, the types of choices faced by its designers are similar to those faced in later and more complex game productions. Figure 5.4 is a table of basic

Perspective	First person		Third person	
Dimensions	Two		Three	
Space type	Torus	Abstract	Free	
Off-screen Space	Dynamic	Static	None	
Scroll	Vertical	Horizontal	Free	None
Exploration	Forced	Free	None	

Figure 5.4 Basic graphical/spatial game characteristics. The figure does not indicate any relationship in the columns

geographical and representational characteristics that we shall use to distinguish between aesthetic strategies in games, followed by a detailed description of each.

Perspective

Excluding text-based games and abstract puzzle games (in which perspective is irrelevant), all video games employ either a first- or a third-person perspective (or the ability to toggle back and forth between the two). In addition, games can also employ either an isometric perspective (similar to an architect's sketch of a building) or a top-down perspective (also known as a bird's-eye perspective).

In the first-person perspective, we see the game action from the point of view of the protagonist—it is as if the player himself is experiencing the action. In the third-person perspective, the player watches whatever it is that he controls, which can be an on-screen object (a character, a vehicle), a number of objects (an army of soldiers, a group of villagers), or various settings (such as tax rates or city zoning). In the case of games like *SimCity*, the player is not even directly represented in the game world, suggesting that we can enjoy interacting with a system without the need (so obviously present in narrative film, for instance) to identify with actual characters. If we consider video games as an extension (or variation) of classical board games, we won't be surprised here. Card games, dice games, or board games such as mancala or backgammon are engrossing without any human-like representation of the player.

What is more surprising is that video games seem to work equally well in both the first and third-person perspectives. Nor is there any indisputable difference between the experience of playing the two. It is easy to assume that the first-person perspective offers a unique, visceral, and hectic experience, but this is belied by the blood-pumping enthusiasm that many players have brought and still bring to even the simplest of third-person action games (like *Spacewar!*).

One general statement we can make, however, is that genres and subgenres (as discussed in Chapter 3) consistently adhere to one or very few

perspectives. For instance, real-time strategy games always employ the third-person perspective. The same holds for turn-based strategy games and for most action games, which require the careful coordination of physical activities by different players. We can imagine the difficulty, for example, of maneuvering for the vine swinging and pyramid climbing of archaeological adventuress Lara Croft while seeing the game world through her eyes.[19] In comparison, almost all modern action games that revolve around shooting are tellingly referred to as first-person shooters; it would be hard to imagine these in the third person. Multiplayer third-person shooters, however, are far less unlikely than a first-person strategy game. The reason for this is that strategy games do not have single protagonists, making such a design experiment rather meaningless.[20]

One subgenre not clearly linked to either perspective is role-playing games (whether single- or multiplayer). Throughout the subgenre's history, both perspectives have been used. In *The Elder Scrolls III: Morrowind*, where the player assumes the role of a single character, she is mainly offered a first-person perspective of events (but can shift to third person). In *Baldur's Gate II: Shadows of Amn*, where the player is responsible for the well-being of an entire party of adventurers, the player must content herself with a third-person perspective. Similarly, MMORPGs usually "prefer" the first-person perspective but always offer a multitude of different perspectives.

Some other games seek the best of both perspectival worlds: they avoid the strict first-person perspective (seeing through the character's eyes) in favor of a point of view placed very close to the object controlled by the player.

As we follow the historical evolution of video game design, it is clear that a strict division between first- and third-person perspectives becomes less helpful; rather, we should discuss a game's point of perception[21]—the point from which the player perceives the gamespace. Importantly, a game may offer varied, often overlapping, points of perception. Real-time strategy games, for instance, offer maps of the entire gamespace that coexist with the standard view chosen by the player.

What makes the difference is often the distance of the point of view from the game action. In *Half-Life 2*, the point of perception is that of the player character, which shows only a very limited portion of the game world as a whole. This ensures very limited knowledge and a frantic, nonstrategic approach to problem-solving. *Baldur's Gate II: Shadows of Amn*, on the other hand, uses the so-called isometric perspective.

The isometric perspective is a technique normally used in architectural drawings, as a method for presenting three-dimensional objects in a two-dimensional form. Technically speaking, the angles between the projection of the three axes (X, Y, and Z) are all 120° and there is no vanishing point as seen in the second *SimCity* game (Figure 5.5). The isometric perspective is not necessarily more effective or engaging than a top-down perspective, but it does allow for a less abstract perception of the gamespace. We can



Figure 5.5 SimCity 2000 (1993) demonstrates the isometric perspective

see this by comparing the top-down perspective of the first *SimCity* game (Figure 4.19) with the sequel *SimCity 2000* (Figure 5.5).

Two early uses of the isometric perspective were the 1982 arcade space action game *Zaxxon* and the 1984 ZX Spectrum game *Knight Lore*, in which a werewolf roamed a magical castle in the hopes of lifting the curse upon him. Notable later games to employ the isometric perspective include *Paperboy* (1985), *Diablo* (1996), and *The Sims* (2000). Today, the perspective is a staple of the real-time strategy subgenre, although early examples—such as *Dune II* and *Warcraft*—in fact used a top-down perspective. Role-playing games with a third-person perspective (such as *Baldur's Gate* or *Lineage: The Bloodpledge*) also always use the isometric perspective, probably because it facilitates identification with on-screen characters and simultaneously allows for a strategic understanding of the gamespace.

Perspective is of great importance in game design, since it directly shapes how we perceive the game world and how close we can get to its characters and objects (and whether we can even relate to individual characters at all).

Dimensions

Although we shall soon discover gray areas, in essence all computer graphics are either two-dimensional or three-dimensional (not to be confused with "3D projection," discussed later). We have already seen examples of both, but now is the time to be more specific.

Two-dimensional graphics are somewhat comparable to paintings. They are created in computer applications like Adobe Photoshop, originally

designed to facilitate drawing, sketching, and image processing. They can be printed on paper without losing essential qualities (unlike 3D graphics which, if printed, would obviously no longer be rotatable, etc.).

Elements of a 2D image may consist of either vector graphics or raster graphics (to name the most important possibilities). Vector graphics are geometric models described as mathematical statements. Thus, they can be turned, twisted, rotated, and resized without any loss of quality. Raster graphics (or bitmap graphics), on the other hand, are rectangular maps of pixels (points of color), each pixel endowed with a color value. When thousands or millions of these individual pixels are grouped together, they create a comprehensible image. Raster graphics are used to transfer photographs from a digital camera onto your computer screen and cannot be resized—at least not upwards—without obvious quality issues.

Three-dimensional graphics are fundamentally different: they add the Z-axis, and 3D graphics objects thus become spatial (see Figure 5.6).

Also, the computer will typically store 3D objects as models (see Figure 5.7), which can then be rendered wherever appropriate in a game, for instance, as running right next to the player character with sunlight illuminating one side. Such on-the-fly rendering can be computationally demanding and thus gaming has tended to drive hardware development.

There are creative ways of achieving 3D-like effects using 2D graphics, to create a feeling of depth without the complexity and cost of real 3D, or simply because it fits the designer's aesthetic purpose. Designers most often use the isometric perspective in this way, as discussed earlier. *SimCity 2000* (Figure 5.5) uses 2D graphics but attains depth by using the isometric perspective.[22] *Wolfenstein 3D*, the first modern-looking first-person shooter, uses a first-person perspective but "fakes" a three-dimensional effect by use of 2D models (see Figure 4.22). Simply put, the enemy soldiers are merely bitmap figures scaled to match their distance from the player's point of view (a visual trick quite obvious if you look for it). These enemy soldiers have no backs—so it's fortunate that they always face the player. A related technique—known as "parallax scrolling"—provides an image with depth by moving layers at different speeds. Thus, objects "close" to the player move faster than more distant objects. This technique is used in *Moon Patrol, Shadow of the Beast*, and *Canabalt*, for instance. Since two-dimensional processes can be used to fake three-dimensionality, distinguishing between whether graphics are technically 2D or 3D is not vital for our discussion of game form. Thus, in this book we categorize

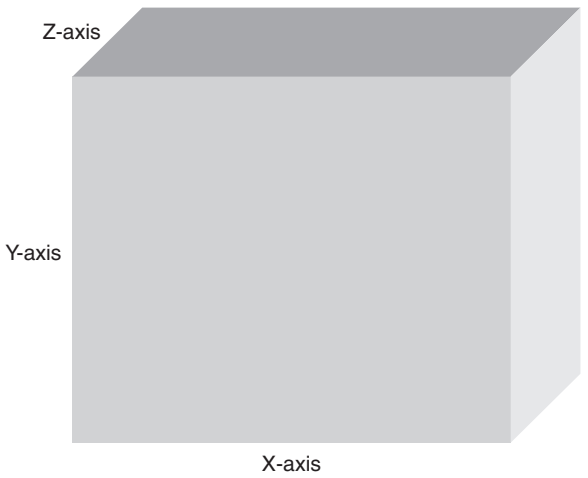


Figure 5.6 In games with three-dimensional graphics, the gamespace can be thought of as the inside of a box

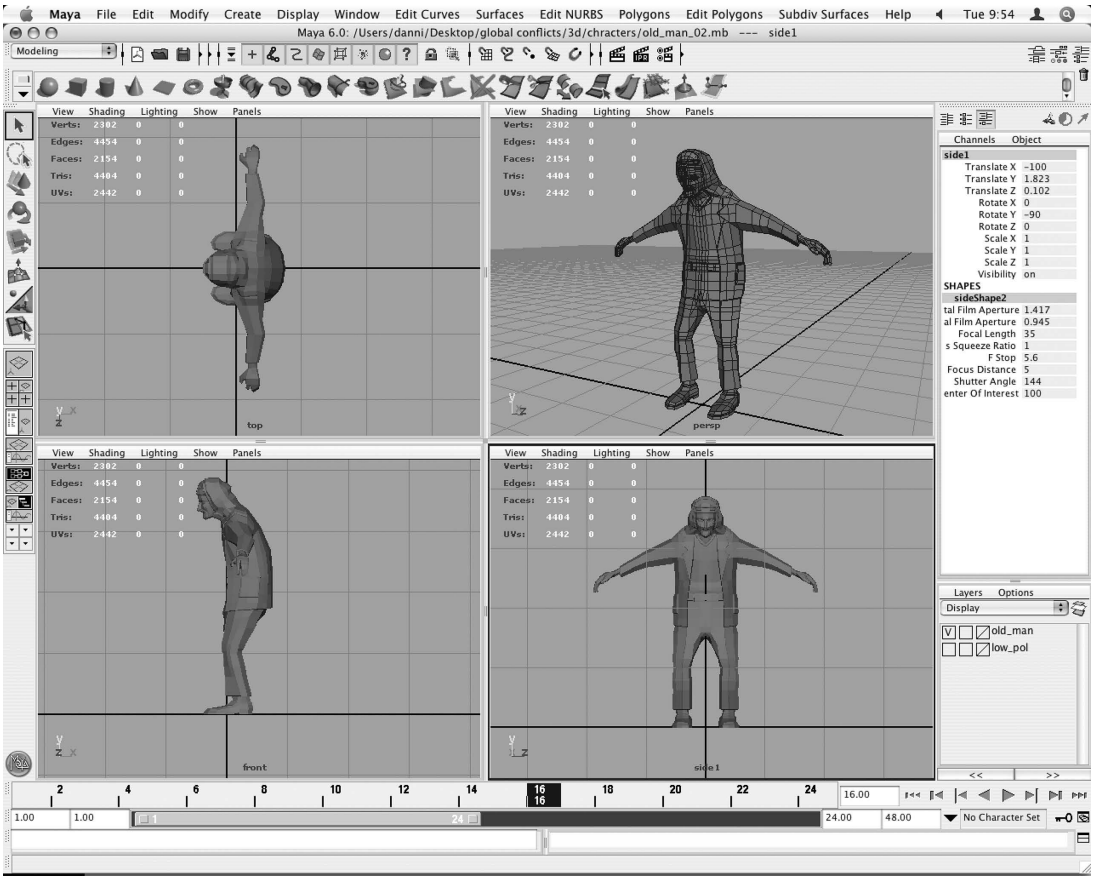


Figure 5.7 Example of 3D model from *Global Conflicts: Palestine* (2007)

as 3D any game that allows for movement in three dimensions—that is, up and down, and into the gamespace.

What, we might ask, is the relationship between a game's perspective and the number of dimensions it uses? Since 3D can be faked using 2D, there seem to be none. While there are no rules about this relationship, nearly all games follow similar standards. A game with third-person perspective can exist in two dimensions or three. But it is hard to imagine a first-person game that does not use (or at least fake) three dimensions. Again, we see how aesthetic aspects that are technically only loosely related tend to be tied together in practice. Creative game design tends to happen through appreciating the existence of such conventions and considering how one may innovatively challenge them.

At this point, it is worth addressing a potential confusion. As most large-scale games have adopted 3D graphics, the term **3D games** is increasingly reserved for games that offer "3D projection." These games can make game objects appear to float between the player and the screen, an experience similar to that of watching a 3D movie. Technically, this can be achieved by

3D games: Refers to gamespaces modeled in three dimensions, or to three-dimensional projection, in which the player's eyes are "tricked" into perceiving depth in the image.

various methods,[23] but it always involves sending a slightly different image to each of the player's eyes (the difference between what is seen by each eye is one reason our brains perceive depth in the real world).[24] The PlayStation 3 (PS3) introduced 3D projection in the spring of 2010, and games like *Gran Turismo 5, MotorStorm: Pacific Rift*, and *Top Spin 4* soon started appearing with differing types of 3D integration. Nintendo made headlines with the early 2011 release of their Nintendo 3DS, a version of the best-selling handheld device offering a 3D experience without the use of special glasses. While early sales were decent, 3D games have not, at the time of writing, taken off conclusively (similar to television with 3D capabilities).

As for 3D games themselves, they have not tended to differ greatly from their 2D versions, although much emphasis has been placed on long objects protruding from the screen.

Space Type

Most nonabstract games of recent vintage occur in a "lifelike" space, comparable to that illustrated in Figure 5.6. But early games like *Spacewar!*, as we have seen, used 2D graphics and had a somewhat idiosyncratic gamespace. When the player-controlled spaceships left the gamespace at one side, they immediately appeared on the opposite side. This gamespace by its nature is abstract. Another interpretation, however, is that the gamespace was in fact circular, or torus-shaped (see Figure 5.8). Thus, going far enough in one direction lands the players back at their starting point.

The *Spacewar!* approach to gamespace was copied by many later games, such as *Pac-Man* and the arcade hit *Defender* (see Figure 4.6). In *Defender*, for no obvious reason, the player navigates a torus (possibly the alien-ridden planet is extremely small and the player merely circles it). This approach has fallen out of fashion in recent decades, as the technological capabilities of game platforms have increased exponentially—and perhaps, in turn, a player's desire for a more "realistic" gamespace to navigate.

And yet modern games, which typically let the player roam more freely, do not allow immediate access to the entire gamespace. We have already discussed how most games limit vertical movement and most also divide the gamespace "box" into various levels that the action may move between.

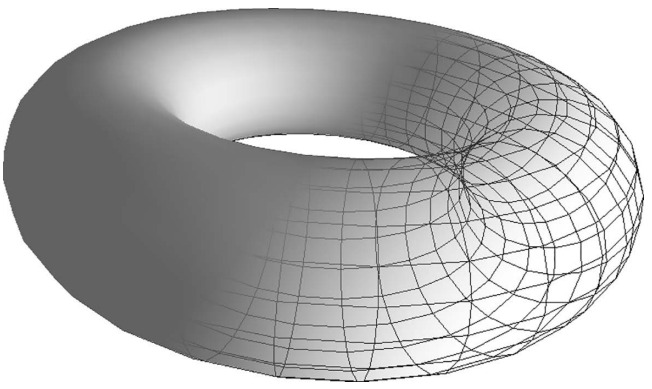


Figure 5.8 The torus illustrates the looping space in some games

An important element of space is the basic size of the gamespace. Early games, like *Spacewar!* and *Pong*, were limited simply by what one screen could hold. But soon games (such as *Night Driver* and *Bump'n'Jump*) started transcending the one-screen gamespace. At least three alternative approaches emerged early on (along with a number

of combinations): unconnected levels, zone-based multiscreen spaces, and seamless multiscreen spaces.

In games with unconnected levels, when a level is cleared the player character is merely transported to a different place; there is not necessarily an implication that the two places are physically connected. Examples of this approach include *Pac-Man* and *Bomb Jack*.

In the case of zone-based multiscreen spaces, the player character exits through one side of the screen and the perspective then jumps to show the player character on the other side of the next screen. For instance, reaching the right side of the screen, the perspective skips to show the player standing on the left side of the next screen, facing right (the implication is that the perspective has moved, not the player character). This type of space was popular with many early graphical adventure games, such as Warren Robbinett's *Adventure*, from 1978.

Finally, seamless multiscreen spaces use a scrolling perspective, so the player gradually reveals the gamespace as he moves. This is the case in games like *Ghosts 'n Goblins, Time Pilot*, and *Super Mario Bros*.

The space type has clear consequences for gameplay. With a one-screen gamespace, all of the objects in a game are usually visible at all times. When all elements are visible, a player may concentrate all resources on that which is known. The player doesn't need to save energy for unexpected situations or use precious mental capacity to plan for various contingencies. The same holds true for games with unconnected levels, although when these levels are linked in terms of the game's goals the player may still have to worry about getting hold of objects needed at a future level, for example, or conserving energy for the final showdown. In seamless multiscreen spaces, quite the opposite is true. Here, projections of what might come must inform the player's choices. In the racing game *Bump'n'Jump*, for instance, the player has to consider what the odds are that various obstacles (such as pools of water) may be looming ahead, and then play accordingly.

Off-Screen Space

The concept of off-screen space originates in cinema. If a movie scene shows two people meeting in an office and the soundtrack contains the sound of traffic in the street below, then that street is part of the scene's off-screen space. In video games, there are two distinct types of off-screen space. We can refer to them as "passive" and "active." In passive off-screen space nothing really happens. This is the case in a game like *Spy Hunter*. The player controls a weapons-laden car as it barrels down an unfolding road, shooting enemy vehicles. We know that the gamespace is larger than what fits in any one screen, and as the road scrolls past we encounter enemies appearing from both below and above. But when they are not visible on the screen, the other vehicles are irrelevant; their behavior and accompanying fate is processed by the game only when they are on-screen. The off-screen space, then, is passive: logically it exists, but nothing happens out there.



Figure 5.9 Spy Hunter (1983) demonstrates well the off-screen space that you know is in front of you but you can't yet see

This is not the case for games in certain other genres. Since real-time strategy games rely heavily on active off-screen space, let's take a look at one representative example—*Warhammer 40,000: Dawn of War*.

As the player's budding warlord lays the foundations of his glorious fortifications on one side of the gamespace, other characters (controlled by additional players or the computer) are doing the same in unseen areas of the game's world. What happens beyond the frame inevitably affects the course of the game. Thus, the off-screen space here is radically different from that of *Spy Hunter*. It is dynamic, living, or active.

The extreme approach to active off-screen space is of course that of MMORPGs, in which the player at any one time experiences only a miniscule portion of the gamespace. In addition, in MMORPGs the game is processed even when the player is not even present in the game world—also referred to as the world being "persistent"—and plays even when the player is not active.

Modern games tend toward more active off-screen space, whereas almost all arcade games had passive off-screen space; but the distinctions are not always clear-cut. Take, for instance, *Grand Theft Auto V*. While often applauded for its "breathing" game world and the openness of its gameplay, *GTA V* is not, in fact, a living simulation of an entire city. At any given moment, objects that are not directly related to the player character (those that are very close or on the screen, or being tracked by the player's radar) are not being processed by the game. As an example: the player hijacks the car of a poor city dweller—the victim does not then (for example) lose his job, become a criminal, and pose a danger to the player in dark alleys. Instead, he or she just disappears when the player character has reached a certain distance from the crime scene. Similarly, a dramatic car crash does not slow down traffic in other places in the city, or leave nearby streets unguarded by police. A similarly partial approach to off-screen space is found in virtually all recent action games, such as *Red Dead Redemption* and *BioShock*. The rarity of an active off-screen world in action games is in sharp contrast to the abundance of life that happens off-screen in MMORPGs and real-time strategy games, and reflects the different priorities of these genres. The former prize the intensity of conflict above all else—the bigger world is quite peripheral—whereas the satisfaction of MMORPGs and real-time strategy

games lies in figuring out the ramifications of the endless possibilities of human (and nonhuman) interaction.

Scroll

As we have seen, "scroll" refers to the gradual unveiling of gamespace. A game can scroll horizontally, vertically, or freely, meaning both vertically and horizontally; or not at all. In scrolling games, the perspective is typically centered on the player character. The concept is not usually applied to first-person games, which do not have an "implied camera" tracking the game action.

Horizontal scroll was common in many arcade games of the 1970s and 1980s. Often, the player character would fight his way from left to right, by either battling or avoiding opponents. Examples of early horizontal scroll abound: *Defender* (1980), *Scramble* (1981), *Kung-Fu Master* (1984), *Ghosts 'n Goblins* (1985), and *Wonder Boy* (1986).

Vertical scroll was slightly less common in early games but could be found in both personal combat-style games and aircraft shooters. Examples include *Spy Hunter* (1983), *1942* (1984), *Commando* (1985), and *Rainbow Islands* (1987). Some arcade games scrolled both ways. Certain games, such as *Time Pilot*, scrolled freely (within the gamespace), while in others, such as *Gauntlet*, movement was constrained by walls or other features.

In the main, it seems that platform games (where the player must jump between platforms) work best with horizontal scroll, one of very few exceptions being *Rainbow Islands*, which did not set an influential precedent (although later games like *New Super Mario Bros.* and *Limbo* include sections of vertical movement).

Scroll is typically only used in the context of 2D graphics, since 3D gamespaces are not usually gradually revealed in the same paper scroll-like manner.

Exploration

In a given game, is the player free to explore the gamespace at her own pace? Or are there constraints—a ticking clock, to name one obvious example? Most arcade games employ the latter principle. Arcade games make more money the more often they're played, so a moving perspective that literally pushes the player forward quickly became the standard. This variation of scroll we call "forced exploration." In the vertically scrolling *Rainbow Islands* mentioned previously, the bottom of the screen slowly filled with water, forcing the player upward. Forced exploration was even more common in vehicle-based shooters, like *1942* or *Scramble*, where the forward propulsion of the airplane (or spaceship) forced the player to keep moving. In sharp contrast, gameplay time is not an issue on consoles or home computers. Here, with the exception of certain flight games, exploration tends to be unforced, and the player is free to navigate as quickly or as slowly as he desires.

Having examined how space is created and used in video games, we turn now to an equally important feature: time.

Time

Time in a video game is also an aesthetic aspect—that is, something experienced by the player. It is also largely under the control of the designer, who may adjust many aspects of time, for example, whether or not it can be sped up during trivial tasks (such as traveling).

Compared to the interest in gamespace, research on the issue of time in games is limited.[25] While time has been a very important topic in literary and film studies, especially from a structural perspective, it draws little attention in game studies. But as Jesper Juul suggests,[26] looking at game temporality can help us understand how different game genres are influenced by different aesthetic conventions, and can even shed some light on the historical development of the industry. If we look at the beginnings of game history, coin-operated arcade machines offered a distinct sense of time compared to today's games: as mentioned previously, arcade games were fast-paced and designed for short playing sessions, in order to maximize the dropping of coins into that bottomless slot. With the arrival of PC games, designers found a very different set of parameters, which allowed for longer playing sessions and encouraged "slow" genres such as adventure games (although such games had been pioneered on earlier machines).

Inspired by concepts of discourse and story time, and how they create a sense of narrative, Juul proposes that we distinguish between play time (the "real" time a player spends playing a game) and event time (the time that passes in the game world during this game). These two concepts apply only to games that project a fictional world; in abstract games, the concept of event time does not make much sense. The relationship (or "mapping") between these two "times" is, Juul suggests, highly variable. For example, in action games such as *Quake* or *Unreal Tournament*, play time and event time are equal, which means that every action the player takes (moving with the mouse, or shooting a bad guy) immediately affects the world inside the game. In this case, we say that the game happens in real time. By contrast, strategy and process-oriented games often compress event time so that action can span across years and even centuries, and a few hours of play can make or destroy civilizations. For example, in *SimCity*, we might play for two minutes of real time (play time), and two years could pass on the game world clock (event time). In some of these games the player can adjust the event time according to their desired speed (typically quicker if they are experienced players and slower if they are still learning how the game works).

Cut-scene: Dramatically important sequence, often displayed without the interaction of the player. The scene is typically shown to motivate a shift in the "plot" of the game and displayed outside of the game engine.

Games that contain **cut-scenes** represent an interesting anomaly: the relation between play time and event time is interrupted, while at the same time cut-scenes contribute to the construction of event time (for example, by showing what happened in an span of five years). Another interruption of play time in many contemporary games is simply a consequence of technology: the player has to pause—for a few seconds or a few minutes—while the console loads the game's many levels. Some game designers consider the inevitable wait a negative player experience. Juul calls these interruptions

of play time "violations." A prime violation, in Juul's opinion, is the much-discussed possibility of saving games. Does the ability to return to a game later on ruin the aesthetic experience of playing? Is it a necessary evil in order to complete difficult games, or is it simply a design feature that can be both used and abused? This is an aesthetic debate among designers, and not one with an obvious answer.

The distinction between play time and event time points to how players understand the mapping between their actions and the game's fictional world, but it does not tell us a lot about the gameplay itself, since in principle every game is played in real time. Even slow adventure games offer an immediate connection between player action (clicking a mouse, for example) and events in the game world (character movement, dialogue, decision-making). The significant difference between play and event time, then, is actually in pace. Think of the reaction speed that different games require of players, and the fact that every genre has conventions of pace and speed that the player must learn to be successful. For example, many of the early (and also abstract) arcade games are about learning to react on time to the same challenges again and again, so that players become quick enough to complete each level.

Time becomes even more significant when we realize that we already categorize several types of games based on how they use time aesthetically. Turn-based games, sports games, and multiplayer action games are all founded upon the particular way that time passes. An even more extreme example is MMORPGs, where a huge time investment is required by the player in order to build a character skilled enough to truly explore the game. Many intriguing questions about time in video games have not yet been explored in detail. Time, however, is such a basic human tool for understanding our place in the world that it continues to be a significant (if subconscious) tool for game designers. Further study of time's many ramifications should only increase our understanding of the building blocks of a video game's design.

Graphical Style

In the previous pages we have glimpsed the variety of elements that a designer must decide between—from perspective to space type to the way time progresses—each of which has obvious implications for gameplay. But designers must also decide on graphical style—how the game should express itself visually. For instance, *Warcraft III* and *Rise of Nations* are both real-time strategy games and share a number of design elements; but to the player they feel rather different, in large part because of their divergent graphical expressions. *Warcraft III* copies a cartoonish graphical style, and makes no pretense of realism, while *Rise of Nations* portrays characters and buildings in a mostly realistic manner.[27] Game scholar Aki Järvinen[28] has identified three graphical styles that have dominated video game design since its inception: photorealism, caricaturism, and abstractionism. We will discuss the three styles briefly in turn.

Photorealism

All media and art forms have at one time or another been employed in the service of realistic representation. In painting, for instance, it was common at least until the mid-nineteenth century for artists to attempt as faithful as possible a rendering of reality on the canvas (the subjects, of course, could be imaginary or fictitious, but the style was realistic). Starting in the 1860s the French impressionists Manet, Renoir, Monet, and others dramatically challenged the dominant style by attending to subjective, fleeting moments (an idea that seems quite conservative, of course, compared to what came after). In the 1960s, following multiple schools of abstraction, came photorealism: a style of painting that tried to mimic photographs. Computer graphics have similarly been unable to resist the lure of the "real." We could argue that video games have attempted to depict reality from their beginnings but were long hampered by insufficient technology. A dominant branch of video game graphics development has craved improved hardware able to process more and more details. Nowhere in video games has such an attempt been more obvious than in the "interactive movies" of the early to mid-1990s (see Chapter 4). Believed by many at the time to be the culmination of games' search for realistic graphical expression, these adventure games typically saw Hollywood actors performing pre-filmed action sequences in filmed environments. Illustrative examples include *The 7th Guest, Ripper, The Beast Within: A Gabriel Knight Mystery*, and *Under a Killing Moon*.

Photorealism has given birth to two subcategories: televisualism and illusionism. The former is a graphical style that attempts faithfully to mimic the aesthetics of television, usually in the form of sports. Such games mimic multicamera productions and use sports TV features such as instant replay. These games copy an evolved form of representation clearly suited to the sport at hand (though not necessarily optimal for a game in terms of control or identification).

Games taking the route of illusionism use photorealistic graphics in the service of nonrealistic content. A science fiction game may attempt to present aliens and unearthly machinery in a fashion that seems lifelike, even though the objects presented have no real-life counterparts. For instance, the looming metallic tripods of *Half-Life 2* seem to move with menacing realism, although the player has no real-life experience of such monstrosities.

Caricaturism

Long a favorite of political cartoonists, caricatures attempt to present the essence of a person or object by exaggerating the most prominent features. In recent years, a few games have taken this approach, often achieving the feeling of a cartoon.

Despite their departure from reality, and their similarity to cartoons, these games are not always intended for younger audiences. Many games based on caricaturism are gripping, which belies the notion that immersion

is intimately linked to realism in any direct sense. *Doodle Jump* and the *Crash Bandicoot* games take a caricaturist approach, as do the *Zelda* and *Jet Set Radio* games.

Doodle Jump was in fact part of a strong trend in mobile casual games in which for a time game designers would aim for a pronounced "sketch aesthetics," a whimsical style invoking the creative process, sometimes making the game look like a draft. This style was employed in games such as *Line Rider* in 2006, *Jelly Car* in 2008, and *Max and the Magic Marker* in 2010 (and some years before, in 1999, in *Vib-Ribbon*).

We can make a few generalizations about the relationship between graphical style and genre. With exceptions like the realistic play of *Jet Set Radio*, caricaturist games are mostly unrealistic (or semi-abstract) in their gameplay. Thus, most platform games fall within this category. Other sub genres—such as **3D shooters** and racing games—cling tightly to photorealism. But many genres resist graphical categories, and contain games in a range of graphical styles. In the case of real-time strategy games, for instance, games such as *Rise of Nations* or *Command & Conquer: Generals* are photorealistic, while *Warcraft III: Reign of Chaos* is caricaturist.

3D shooter: Action games in which the action is seen through the eyes of the protagonist and where the graphics are three-dimensional (and often constructed of polygons). Synonym: first-person shooter (**FPS**).

Abstractionism

Our third category is a visual style that does not try to represent people or real-life objects. Though rarely seen in commercial titles, it is far from unimportant in the industry's continuous experiments with game form. At heart, abstractionism can be said to be "about" form.[29] A prime example

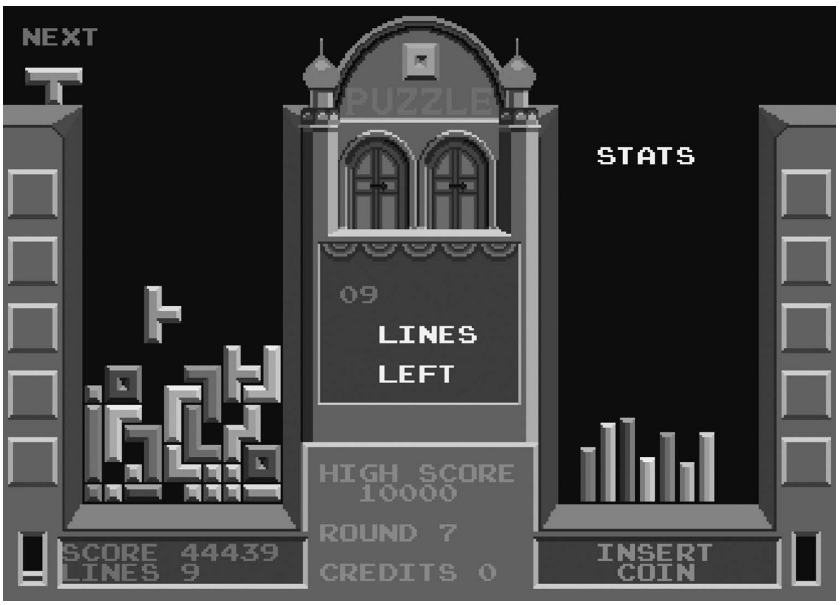


Figure 5.10 *Tetris* (1987) is world famous with numerous permutations and is among the most successful abstract games

here is the legendary Soviet smash hit *Tetris*, which began its worldwide reign in the mid-1980s.

Long before *Tetris*, however, there was 1981's *Qix*, in which the player draws rectangular shapes while avoiding Qix and Sparx, the game's abstract, line-drawn enemies.[30] The same year, and more famously, there was *Tempest*, in which a yellow, three-dimensional crab-creature valiantly defended a barely delineated world from strange abstract beasts that continually emerged from a tunnel. More recently, abstractionism saw a brief resurgence of interest with the courageous *Rez*, a rhythm-based shooter game in which the player traverses a psychedelic gamespace populated by colorful shapes and figures. While scoring big with critics and the experimentally minded, *Rez* was less of a commercial success.[31] The game's lack of influence led Aki Järvinen to worry that "abstractionism might end up as a marginal [footnote] in the history, and future, of an audiovisual cultural form also known as computer and video games."[32]

The overarching problem with abstract games, not surprisingly, is that they are hard to market. It is difficult to create hype purely around a game's mechanics (as opposed to its amazing story, or its sexy lead character) and next to impossible to make money on merchandise (not to mention movie rights), while a trailer of an abstract game is unlikely to wow the marketers and buyers at game fairs. Equally important is the fact that contemporary games seem to attach increasing importance to narrative; just as film and literature have proven over time, despite ongoing murmurs of experimentation, humans seem ineluctably drawn to the simplest form of all—the story. But while AAA titles will rarely be fully abstract, mobile gaming has breathed life into tile matching and *Tetris*-like puzzle games, and so abstractionism thrives in your pocket with titles such as *Drop7, Glow Puzzle*, and *Blue Block*.

Game Audio

Think about your favorite game. Now try to describe what it sounds like. Unless you are unusually analytical about your video games, you will likely be (a) unable to remember the sound and music clearly or (b) unable to analyze the sound design in any detail. When we want to approach game sound systematically, we face two problems. One is a broad, and embarrassing, tradition of ignoring the audio side of audiovisual media. In film studies, for instance, sound has received extremely limited attention relative to its importance to the experience of a movie.[33] The second problem is that we are generally not taught how to describe the qualities of sound. In fact, our common vocabulary does not even include many terms that can be used to differentiate sound types. This causes a problem for interaction designers—professionals who design interfaces in the widest sense—who want users to state their sound preferences or comment on sound design. To comment systematically on this important element of games, one must become familiar with a complex language ("transparent," "narrow," and "full," for example, are adjectives used to describe sounds).